An LSTM-Based Fully Convolutional Network for Pedestrian Intersection Estimation and Intention Tracking in Crowded Zones

Kalyan K. Parameshwaran[1] Shinkyu Park[1]
1 Visual Computing Center / Robotics, Sensing and Control Systems Group,
King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia
Visiting Student Research Program (VSRP) · 2021

Abstract

Predicting whether and when a pedestrian will cross a roadway is a central perception task for autonomous driving and intelligent crowd-monitoring systems, and it becomes markedly more difficult in dense urban intersections, where occlusion, elevated pedestrian density, and heterogeneous motion patterns confound classical tracking-based approaches. This paper proposes an LSTM-based Fully Convolutional Network (LSTM-FCN) that jointly performs intersection estimation, identifying the crosswalk region a pedestrian is most likely to enter, and intention tracking, continuously updating a crossing/not-crossing probability as new observations arrive. The architecture couples a fully convolutional spatial encoder, which extracts appearance and local-context features from each frame without discarding spatial resolution through early fully connected bottlenecks, with a recurrent LSTM module that models the temporal evolution of pedestrian pose, trajectory, and scene context. A two-head output design decouples the discrete intersection-zone classification task from the continuous intention-tracking task while sharing a common spatio-temporal backbone, a design choice we argue is beneficial for sample efficiency in crowded scenes. We present a complete experimental protocol for benchmarking the proposed model on the JAAD and PIE pedestrian intention datasets against four representative baselines, a graph neural network model in the style of CVPR-GNN, a convolutional social-pooling model (SocialCNN), and an AlexNet-based per-frame classifier, using standard classification and trajectory metrics. This manuscript documents the complete architecture, training methodology, and evaluation protocol in full; quantitative results are reported in template form (Sections 6–7) pending completion of the experimental runs at KAUST.

Keywords: pedestrian intention prediction; fully convolutional networks; long short-term memory; intersection estimation; crowded-scene understanding; autonomous driving perception; multi-task learning; JAAD; PIE.

1. Introduction

Urban intersections are among the most safety-critical environments for both autonomous vehicles and pedestrian-monitoring infrastructure. In these zones, pedestrians frequently exhibit ambiguous, rapidly changing behavior, approaching a curb, hesitating, glancing at oncoming traffic, and either committing to a crossing or retreating, within a span of one to two seconds. Autonomous driving stacks and smart-intersection systems must therefore not only detect and track pedestrians, but continuously reason about their short-horizon intent: will this pedestrian enter the roadway, and if so, through which region of the crosswalk?

We refer to the first sub-problem as intersection estimation: given a short observation window, predict which discrete crossing zone (e.g., near-side crosswalk, far-side crosswalk, mid-block gap) a pedestrian is most likely to use. We refer to the second sub-problem as intention tracking: given a continuously growing observation stream, maintain an up-to-date probability that the pedestrian is currently in a crossing state. These tasks are related but formally distinct: intersection estimation is fundamentally a spatial classification problem conditioned on scene geometry, whereas intention tracking is a temporal state-estimation problem conditioned on pose and motion history. Treating the two as a single, undifferentiated prediction target, as much prior literature implicitly does, obscures this distinction and, we argue, leaves representational capacity on the table that a two-head formulation can recover.

Crowded zones amplify the difficulty of both tasks. Dense pedestrian traffic causes frequent partial and full occlusions, and social interactions between nearby pedestrians (e.g., group crossings, yielding behavior) introduce dependencies that single-agent motion models cannot capture. Prior work has approached pedestrian intention prediction from three broad angles: (i) recurrent models over trajectory and pose keypoints, (ii) convolutional models over cropped pedestrian appearance and local context, and (iii) graph-based models that explicitly encode pedestrian-pedestrian and pedestrian-scene interactions. Each family has complementary strengths and weaknesses: recurrent models capture temporal dynamics well but are often shallow in their spatial feature extraction; convolutional models capture rich appearance cues but are typically applied per frame without an explicit temporal state; and graph-based models capture social context effectively but add architectural and computational complexity that may be difficult to justify outside of the densest scenes.

This paper proposes an LSTM-based Fully Convolutional Network (LSTM-FCN) designed to combine the spatial feature richness of fully convolutional encoders with the temporal modeling capacity of recurrent networks, while remaining substantially lighter than graph-based alternatives. Our design choice to use a fully convolutional encoder, as opposed to a convolutional-then-fully-connected encoder, is motivated by the observation that spatial resolution in a pedestrian's local neighborhood, curb position, crosswalk markings, and nearby pedestrians and vehicles, carries information that is degraded by early flattening into a fixed-length vector. The convolutional feature map is instead pooled and fused with a trajectory/pose embedding immediately before being passed into the temporal module, preserving spatial structure for as long as is practical within the pipeline.

The contributions of this work are fourfold. First, we propose a two-head LSTM-FCN architecture that shares a spatio-temporal backbone between the intersection-estimation and intention-tracking tasks. Second, we introduce a fusion mechanism that combines fully convolutional scene features with pose and trajectory embeddings prior to temporal encoding. Third, we specify a complete benchmarking protocol against a GNN-style baseline, SocialCNN, and an AlexNet-based classifier on the JAAD and PIE datasets, including the metrics, splits, and training configuration required to reproduce the comparison. Fourth, we provide an extended discussion of the architectural trade-offs relevant to deployment in crowded, occlusion-heavy intersection scenes, together with a stratified evaluation protocol intended to isolate where the proposed architecture is expected to provide the greatest benefit.

The remainder of this manuscript is organized as follows. Section 2 surveys related work on pedestrian intention datasets and modeling paradigms. Section 3 formalizes the joint intersection-estimation and intention-tracking problem. Section 4 details the proposed LSTM-FCN architecture. Section 5 describes the experimental protocol, baselines, and metrics. Sections 6 and 7 present the results and ablation study in template form. Section 8 discusses the architectural implications of the proposed design, Section 9 addresses limitations and ethical considerations, and Section 10 concludes with directions for future work.

2. Related Work

2.1 Pedestrian Intention Datasets

The Joint Attention in Autonomous Driving (JAAD) dataset and the Pedestrian Intention Estimation (PIE) dataset are the two most widely used benchmarks for pedestrian crossing-intention research. JAAD provides short annotated clips of pedestrian-vehicle interactions at crossings, with behavioral tags such as looking, walking, and crossing state. PIE extends this setting with longer, continuous driving sequences, ego-vehicle odometry, and explicit binary crossing/not-crossing labels aligned to a fixed future horizon, making it well suited to the intention-tracking formulation used in this paper. Both datasets are treated here as evaluation benchmarks: the models compared in Section 5 are all trained and tested on the same JAAD/PIE splits, rather than being alternative datasets compared against one another.

A recurring critique of both datasets, which we adopt as a design constraint rather than dismiss, is that annotated crossing events are comparatively rare relative to non-crossing frames, producing a class-imbalance problem that any intention-tracking model must explicitly address. We return to this point in Section 4.5 and in the stratification protocol proposed in Section 8.

2.2 Convolutional and Recurrent Approaches

Early pedestrian intention models applied 2D convolutional neural networks, including AlexNet-style architectures, to cropped pedestrian bounding boxes on a per-frame basis, with predictions aggregated across a sliding window. These approaches benefit from strong appearance features but are limited in their capacity to model how intention evolves over time, since temporal reasoning is typically bolted on as a post-hoc voting or smoothing step rather than being learned end-to-end. Subsequent work introduced recurrent layers (LSTM or

GRU) over pose keypoints, bounding-box trajectories, and CNN feature sequences, improving temporal consistency at the cost of relying on comparatively shallow spatial encoders. This trade-off, temporal depth purchased at the expense of spatial resolution, motivates the fully convolutional design adopted in the present work, since crowded scenes are precisely the setting in which fine-grained spatial context (e.g., which of several nearby pedestrians is closest to the curb) is most informative.

2.3 Graph Neural Network and Social Context Models

A separate line of work models pedestrian behavior with graph neural networks, treating each pedestrian, and in some formulations nearby vehicles, as a node and learning message-passing functions over edges that represent spatial proximity or social interaction. We refer to this family generically as CVPR-GNN-style models in this paper, reflecting the graph-based interaction-modeling approach that has appeared in several CVPR-venue pedestrian trajectory and intention papers. SocialCNN architectures instead encode social context implicitly, using convolutional social-pooling grids that aggregate neighboring pedestrians' state into a fixed-size spatial tensor prior to convolutional processing. Both families explicitly target crowded-scene interactions, which motivates their inclusion as baselines in a crowded-intersection benchmark.

2.4 Attention and Transformer-Based Extensions

More recent work in trajectory and intention prediction has explored self-attention as an alternative to recurrence for modeling temporal dependencies, and as an alternative to convolutional social pooling for modeling inter-agent dependencies. While such mechanisms are architecturally attractive, in that they permit direct, learned weighting of relevant past frames or nearby agents without the fixed-window or fixed-radius assumptions of convolutional pooling, they typically require substantially larger training corpora to realize their benefit over recurrent baselines. Given the comparatively modest scale of JAAD and PIE relative to the datasets on which attention-based trajectory models have been shown to outperform recurrent baselines, we position attention-based extensions as a natural direction for future work (Section 10) rather than as a baseline in the present protocol, and we return to this point when discussing the design space in Section 2.5.

2.5 Positioning of This Work

Relative to purely recurrent, purely convolutional, and graph-based baselines, the LSTM-FCN proposed here occupies a deliberate middle ground along two axes, spatial resolution and interaction modeling, that we use throughout this paper to organize the comparison. It retains full spatial resolution through a fully convolutional backbone, unlike shallow-CNN-plus-LSTM hybrids; it avoids the need for explicit graph construction and message passing over neighboring agents, unlike CVPR-GNN-style models; and it models temporal dynamics explicitly through an LSTM rather than a fixed convolutional pooling window, unlike SocialCNN. The two-head design further distinguishes this work from prior single-task formulations by explicitly separating the discrete intersection-estimation objective from the continuous intention-tracking objective while sharing representation learning between them, a form of soft inductive bias that we hypothesize improves sample efficiency in the comparatively small, imbalanced regime characteristic of JAAD and PIE.

3. Problem Formulation

Let a scene observation at time t consist of an RGB frame crop around a tracked pedestrian, $x_t \in \mathbb{R}^{H\times W\times 3}$, together with a pose/trajectory vector $p_t \in \mathbb{R}^d$ containing bounding-box position and size, body-keypoint coordinates, and ego-motion features when available. Given an observation window $X_{(t-T:t)} = \{(x_i, p_i)\}_{i=t-T}^{t}$ of length T, the model must produce two outputs.

> Intersection estimation: a categorical distribution $\hat{y}_{zone} = \mathrm{softmax}(g(X_{(t-T:t)}))$ over K discrete crossing-zone classes (e.g., near-side crosswalk, far-side crosswalk, mid-block gap, no imminent crossing).
>
> Intention tracking: a per-frame scalar $\hat{y}_{cross}(t) = \sigma(h(X_{(t-T:t)})) \in [0,1]$ estimating the probability that the pedestrian is in, or about to enter, a crossing state at time t, updated as the observation window slides forward.

Both $g(\cdot)$ and $h(\cdot)$ are implemented as separate output heads on top of a shared spatio-temporal encoder $f(\cdot)$, described in Section 4, such that $\hat{y}_{zone} = \mathrm{softmax}(g(f(X_{(t-T:t)})))$ and $\hat{y}_{cross}(t) = \sigma(h(f(X_{(t-T:t)})))$. This decomposition makes explicit the sense in which the two tasks are coupled only through the shared representation

$f(\cdot)$ and are otherwise free to specialize through the independent heads $g(\cdot)$ and $h(\cdot)$, a property we exploit directly in the ablation protocol of Section 7.

4. Proposed LSTM-FCN Architecture

Figure 1 shows the overall architecture. The network consists of four stages: (i) a fully convolutional spatial encoder applied to each frame in the observation window, (ii) a trajectory/pose embedding branch, (iii) a spatio-temporal fusion module that combines the two, followed by an LSTM temporal encoder, and (iv) two task-specific output heads. Each stage is described in turn below, followed by the joint training objective that couples them.

4.1 Fully Convolutional Spatial Encoder

Each frame crop x_t is passed through three convolutional blocks (Conv–BatchNorm–ReLU, kernel sizes 7, 5, 3, strides 2, 2, 1), producing a spatial feature map $F_t \in \mathbb{R}^{(H' \times W' \times C)}$ with $C = 256$ channels. Critically, no fully connected bottleneck is applied at this stage: spatial resolution is preserved through global average pooling only at the point of fusion (Section 4.3), which we hypothesize preserves more usable context for crowded scenes than early flattening into a fixed-length vector. Dilated convolutions with dilation rate 2 are used in the third block to enlarge the receptive field around the pedestrian without further downsampling, which is intended to help the encoder capture nearby pedestrians and vehicles relevant to a crossing decision without incurring the resolution loss of additional strided convolutions.

4.2 Trajectory and Pose Embedding

In parallel, bounding-box trajectory and body-keypoint sequences are embedded with a two-layer multilayer perceptron ($128 \rightarrow 128$ units, ReLU activations) applied independently at each timestep, producing $\tilde{p}_t \in \mathbb{R}^{128}$. This branch is deliberately lightweight: its role is to inject precise, low-noise motion cues, velocity, heading change, and torso orientation, that are comparatively easy to extract from tracker and pose-estimator outputs but comparatively hard for a convolutional encoder to infer implicitly from raw pixels, particularly under partial occlusion. We view this branch as complementary rather than redundant with the visual encoder, and we test this assumption directly in ablation (b) of Section 7.

4.3 Spatio-Temporal Fusion and LSTM Encoder

The pooled spatial feature vector and the trajectory embedding are concatenated and passed through a 1×1 convolution, equivalently a linear projection, to a shared 256-dimensional fusion vector z_t at every timestep. The sequence $z_{(t-T:t)}$ is then fed into a two-layer LSTM with hidden size 256, following the standard LSTM recurrence $h_t, c_t = \mathrm{LSTM}(z_t, h_{(t-1)}, c_{(t-1)})$. The final hidden state h_t summarizes the spatio-temporal history of the pedestrian and local scene over the observation window and is shared between both output heads, described next. We note that because h_t is recomputed as the window slides forward, the intention-tracking head in particular benefits from a form of implicit smoothing across overlapping windows, which we discuss further in Section 8.

4.4 Intersection Estimation Head

The intersection-estimation head is a two-layer MLP ($256 \rightarrow 128 \rightarrow K$) applied to h_t, followed by a softmax over the K crossing-zone classes. It is trained with standard categorical cross-entropy against the ground-truth crossing zone associated with each clip. Because zone labels are assigned at the clip level rather than the frame level, this head is trained on the final hidden state of each observation window rather than at every timestep, in contrast to the intention-tracking head described next.

4.5 Intention Tracking Head

The intention-tracking head is a single-layer MLP ($256 \rightarrow 1$) applied to h_t at every timestep, followed by a sigmoid, producing a running crossing probability. It is trained with binary cross-entropy against per-frame crossing/not-crossing labels, optionally reweighted to account for class imbalance between crossing and non-crossing frames, which is typically severe, since crossing frames are a minority class in both JAAD and PIE. We adopt inverse-frequency class weighting as the default reweighting scheme, with the weighting coefficient itself treated as a tunable hyperparameter alongside the multi-task loss weights described in Section 4.6.

4.6 Joint Training Objective

The two heads are trained jointly with a weighted multi-task loss, $L = \lambda_{zone} \cdot L_{CE}(\hat{y}_{zone}, y_{zone}) + \lambda_{cross} \cdot L_{BCE}(\hat{y}_{cross}, y_{cross})$, with loss weights λ_{zone} and λ_{cross} tuned on a held-out validation split (Section 5.4). Sharing the backbone between the two losses is intended to regularize the spatial and temporal representations: the zone-classification objective encourages the encoder to retain scene-geometry cues, while the intention-tracking objective encourages sensitivity to short-term pose and motion changes. We treat the degree to which these two objectives are complementary, rather than competing, as an empirical question rather than a foregone conclusion, and we designed ablation (d) in Section 7 specifically to test it.

5. Experimental Setup

5.1 Datasets

JAAD is used for intersection-estimation evaluation and for pretraining the spatial encoder, using the standard train/validation/test video-level split to avoid frame leakage across sets. PIE is used for intention-tracking evaluation given its continuous, longer-horizon annotations and explicit future crossing labels aligned to a fixed prediction horizon, commonly evaluated at 0.5 s, 1.0 s, and 1.5 s look-ahead. Both datasets are restricted to clips containing multiple simultaneously visible pedestrians, to emphasize the crowded-zone setting targeted by this paper; single-pedestrian clips are retained only for the intersection-estimation pretraining stage, where the larger effective sample size is expected to improve the stability of the fully convolutional encoder prior to fine-tuning on crowded clips.

5.2 Baselines

CVPR-GNN — a graph neural network baseline following the pedestrian-vehicle interaction graph formulation common to CVPR-venue trajectory and intention papers, with pedestrians and nearby agents as nodes and learned edge message passing.

SocialCNN — a convolutional social-pooling baseline that aggregates neighboring pedestrian states into a spatial occupancy grid prior to CNN processing.

AlexNet — a per-frame AlexNet classifier over cropped pedestrian appearance, with temporal aggregation via majority voting over the observation window, representing a purely convolutional, non-recurrent baseline.

LSTM-FCN — the architecture described in Section 4.

All baselines are re-implemented rather than run through third-party checkpoints, so that differences in reported performance can be attributed to architecture rather than to incidental differences in preprocessing, augmentation, or optimizer configuration, as detailed in Section 5.4.

5.3 Metrics

Intersection estimation is scored with top-1 accuracy, macro-F1 across the K crossing-zone classes, and a confusion matrix used to characterize systematic zone-adjacency errors, that is, cases where the model predicts a crossing zone spatially adjacent to the ground-truth zone rather than an unrelated one. Intention tracking is scored with AUC, F1 at a probability threshold of 0.5, and time-to-event accuracy, defined as the mean absolute difference, in frames, between the predicted and ground-truth crossing onset time for clips in which a crossing occurs. We additionally report inference latency per frame and parameter count for each model, to characterize the accuracy/efficiency trade-off relevant to onboard deployment on resource-constrained platforms.

5.4 Implementation Details

All models are trained with the Adam optimizer, an initial learning rate of 1×10^{-4} with cosine decay, batch size 32, and early stopping on validation macro-F1 with patience 10. The observation window length T is set to 16 frames, approximately 1 s at 15 fps, matching the temporal resolution of the JAAD/PIE annotations. Data augmentation includes random horizontal flipping, with corresponding label adjustment for left/right crossing zones, color jitter, and random temporal cropping. Loss weights λ_{zone} and λ_{cross} are selected via grid search over {0.5, 1.0, 1.5, 2.0} on the validation split. All baselines are re-implemented and trained under an identical data pipeline, augmentation policy, and compute budget as the proposed model, to ensure a fair, controlled comparison in which observed performance differences can be attributed to architecture.

To support statistically defensible comparisons once training is complete, we additionally specify that each configuration will be trained under five random seeds, with reported metrics given as mean ± standard deviation, and pairwise differences between LSTM-FCN and each baseline assessed with a paired significance test at the clip level. This protocol is recorded here, in advance of running the experiments, so that the eventual comparison is pre-registered against a fixed evaluation plan rather than selected post hoc.

6. Results

This section reports benchmark comparisons in template form. The table structure, metrics, and baselines match the protocol of Section 5; the numerical entries below are placeholders included to specify the exact format in which results will be reported, and are to be populated once the training runs described above are completed at KAUST.

Model	JAAD Zone Acc. (%)	JAAD Macro-F1	PIE Intent AUC	PIE Intent F1	Params (M)	Latency (ms)
AlexNet (per-frame)	71.2	0.64	0.72	0.58	60.0	8
SocialCNN	76.5	0.70	0.81	0.68	18.4	14
CVPR-GNN	78.9	0.73	0.84	0.71	22.7	27
LSTM-FCN	80.3	0.75	0.86	0.74	15.2	16

Table 1. Benchmark comparison template on JAAD (intersection estimation) and PIE (intention tracking).

We expect, based on the architectural analysis in Section 4, the following qualitative pattern once results are populated. AlexNet should trail the other models on temporal metrics, PIE Intent AUC and F1, owing to its lack of explicit temporal modeling. SocialCNN and CVPR-GNN should perform competitively on scenes with dense pedestrian interaction owing to their explicit social-context modeling, with CVPR-GNN likely carrying a latency penalty attributable to graph construction and message passing. LSTM-FCN is hypothesized to be competitive on accuracy while retaining lower latency than CVPR-GNN, owing to its convolutional, rather than graph-based, treatment of spatial context. We emphasize that these are hypotheses to be confirmed or refuted by the completed experiments described in this protocol, and not reported findings.

7. Ablation Study

To isolate the contribution of each architectural component, we propose the following ablation protocol, again reported here as a template pending experimental completion. Ablation (a) removes the fully convolutional encoder in favor of a standard CNN-plus-global-pool-plus-FC encoder, to test the value of preserved spatial resolution. Ablation (b) removes the trajectory/pose embedding branch, to test the value of explicit motion cues relative to purely appearance-derived motion. Ablation (c) replaces the LSTM with a fixed-window temporal average, to test the value of learned temporal state relative to a non-parametric aggregation. Ablation (d) trains the two heads independently rather than jointly, to test the value of shared representation learning described in Section 4.6.

Configuration	PIE Intent AUC	Δ vs. Full Model
Full LSTM-FCN	0.86	N/A
(a) w/o fully conv. encoder (CNN+FC)	0.81	−0.05
(b) w/o pose/trajectory branch	0.83	−0.03
(c) LSTM → fixed-window average	0.80	−0.06
(d) independently trained heads	0.84	−0.02

Table 2. Ablation study template isolating the contribution of each architectural component.

If the completed experiments confirm the pattern shown above, the largest expected contribution comes from the LSTM temporal encoder (ablation c), followed by the fully convolutional spatial encoder (ablation a), with the pose/trajectory branch and joint training contributing smaller, but non-negligible, improvements. We treat this ordering as a hypothesis rather than a conclusion, consistent with the template status of all numerical results in this manuscript, and we discuss how the ablation results, once obtained, should be interpreted relative to the discussion in Section 8.

8. Discussion

The architectural motivation behind LSTM-FCN centers on a specific hypothesis about crowded intersection scenes: that the marginal value of preserved spatial resolution, from the fully convolutional encoder, and explicit motion cues, from the pose/trajectory branch, is highest precisely when scenes are visually cluttered and pedestrians are frequently partially occluded. In sparse scenes, we would expect the gap between LSTM-FCN and simpler baselines such as AlexNet-with-voting to narrow, since there is less contextual information available for the additional architectural capacity to exploit. This suggests that evaluation should be stratified by pedestrian density, rather than reported only in aggregate over JAAD and PIE, in order to properly characterize the operating regime in which the proposed architecture provides the most benefit. We propose that any completed evaluation report at least two density strata, low-density clips with at most two simultaneously visible pedestrians and high-density clips with more than two, computed separately for each of the metrics in Section 5.3.

A second discussion point concerns the choice to share a backbone between the intersection-estimation and intention-tracking heads. This design assumes positive transfer between the two tasks, that is, that learning to localize crossing zones improves the features available for tracking crossing intent, and vice versa. This assumption should be validated empirically, via Table 2, row (d), rather than taken as given, since multi-task sharing can also produce negative transfer when the two objectives pull the shared representation in conflicting directions, for example if zone classification rewards static scene-layout features while intention tracking rewards dynamic pose-change features. Should ablation (d) show a small or negligible gap between joint and independent training, this would argue against the shared-backbone hypothesis and would motivate revisiting the fusion design in Section 4.3.

Finally, the choice of CVPR-GNN, SocialCNN, and AlexNet as baselines is intended to span the design space along two axes: temporal modeling, absent in AlexNet, implicit windowing in SocialCNN, and explicit recurrence in LSTM-FCN, and social or interaction modeling, absent in AlexNet and in LSTM-FCN's per-pedestrian branch, convolutional pooling in SocialCNN, and explicit graph structure in CVPR-GNN. A finding that LSTM-FCN underperforms CVPR-GNN specifically in high-density clips, despite competitive performance in low-density clips, would constitute informative evidence that explicit graph-based interaction modeling is necessary in crowded zones, and would motivate a follow-up architecture that adds a lightweight interaction module to LSTM-FCN, a direction we outline further in Section 10.

A further consideration, not addressed in the original protocol, concerns calibration of the intention-tracking output. Because $\hat{y}\_cross(t)$ is intended to inform downstream safety-relevant decisions, we consider it insufficient to report discrimination metrics such as AUC alone; a completed evaluation should additionally report a calibration curve and expected calibration error, since a model that ranks crossing and non-crossing frames correctly but is systematically over- or under-confident could still produce unsafe downstream behavior when its output is thresholded or fed into a planning module.

9. Limitations and Ethical Considerations

This manuscript presents architecture and protocol; the benchmark comparison in Sections 6 and 7 is not yet backed by completed experimental runs and should not be cited as reporting real performance figures until updated with actual results. More generally, pedestrian intention models trained on JAAD and PIE inherit the geographic and demographic composition of those datasets, predominantly North American and European urban driving footage, and performance may not transfer directly to other traffic cultures, lighting conditions, or crosswalk designs without additional data or domain adaptation.

Because the model outputs are intended to inform safety-relevant decisions in autonomous driving or crowd-monitoring systems, any deployment should include conservative fallback behavior for low-confidence predictions, human oversight during validation, and explicit testing for performance disparities across pedestrian subgroups, for example children, wheelchair users, and individuals with strollers, who may exhibit motion patterns underrepresented in the training data. We further note that any deployment of a crowd-monitoring variant of this system raises surveillance considerations distinct from the in-vehicle perception use case, and we recommend that such deployments be subject to separate privacy and data-governance review rather than treated as a direct extension of the autonomous-driving setting.

10. Conclusion and Future Work

We presented LSTM-FCN, a two-head architecture for joint pedestrian intersection estimation and intention tracking that combines a fully convolutional spatial encoder, a pose/trajectory embedding branch, and an LSTM temporal encoder. We described a complete benchmarking protocol against CVPR-GNN, SocialCNN, and AlexNet baselines on the JAAD and PIE datasets, including metrics, training configuration, and an ablation plan for isolating the contribution of each architectural component.

Future work includes, first, completing the experimental runs described in Sections 5 through 7 and reporting the results with the significance-testing protocol specified in Section 5.4; second, extending the interaction modeling in LSTM-FCN with a lightweight graph or attention module to better capture pedestrian-pedestrian dependencies in dense crowds, informed by where CVPR-GNN is found to outperform the proposed model; third, evaluating cross-dataset generalization by training on JAAD/PIE and testing on an independent intersection dataset collected at KAUST or a partner benchmark; and fourth, incorporating the calibration analysis proposed in Section 8 as a standard component of the evaluation protocol, given the safety-relevant nature of the intended deployment setting.

Acknowledgments

K. K. Parameshwaran completed this work as a Visiting Student Research Program (VSRP) internship at KAUST under the supervision of S. Park.

References

[1] A. Rasouli, I. Kotseruba, and J. K. Tsotsos, "Agreeing to cross: How drivers and pedestrians communicate," in IEEE Intelligent Vehicles Symposium (IV), 2017. (JAAD dataset)

[2] A. Rasouli, I. Kotseruba, T. Kunic, and J. K. Tsotsos, "PIE: A large-scale dataset and models for pedestrian intention estimation and trajectory prediction," in IEEE/CVF International Conference on Computer Vision (ICCV), 2019.

[3] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[4] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in Advances in Neural Information Processing Systems (NeurIPS), 2012. (AlexNet)

[5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[6] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2015.

[7] A. Vemula, K. Muelling, and J. Oh, "Social attention: Modeling attention in human crowds," in IEEE International Conference on Robotics and Automation (ICRA), 2018.

[8] N. Rhinehart, R. McAllister, K. Kitani, and S. Levine, "PRECOG: Prediction conditioned on goals in visual multi-agent settings," in IEEE/CVF International Conference on Computer Vision (ICCV), 2019.

[9] I. Kotseruba, A. Rasouli, and J. K. Tsotsos, "Benchmark for evaluating pedestrian action prediction," in IEEE Winter Conference on Applications of Computer Vision (WACV), 2021.

[10] K. Saleh, M. Hossny, and S. Nahavandi, "Spatio-temporal DenseNet for real-time intent prediction of pedestrians in urban traffic environments," Neurocomputing, 2019.

[11] S. Yagi, K. Mangalam, R. Yonetani, and Y. Sato, "Future person localization in first-person videos," in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018.

[12] A. Vaswani et al., "Attention is all you need," in Advances in Neural Information Processing Systems (NeurIPS), 2017.

Appendix A: Notation Summary

Symbol	Meaning
x_t	RGB frame crop of tracked pedestrian at time t
p_t	Pose / trajectory feature vector at time t
T	Observation window length (frames)
F_t	Fully convolutional spatial feature map at time t
p̃_t	Trajectory/pose embedding at time t
z_t	Fused spatio-temporal feature vector at time t
h_t, c_t	LSTM hidden and cell states at time t
K	Number of discrete crossing-zone classes
ŷ_zone	Predicted crossing-zone distribution
ŷ_cross(t)	Predicted crossing probability at time t
λ_zone, λ_cross	Multi-task loss weights

Appendix B: Reproducibility Checklist

In line with standard reproducibility practice for empirical computer vision papers, the following items will accompany the completed experimental release of this work: (1) exact train/validation/test video IDs used for the JAAD and PIE splits; (2) full hyperparameter configuration files for LSTM-FCN and all three baselines; (3) random seeds and the number of repeated runs used to compute reported means and standard deviations, per the protocol in Section 5.4; (4) hardware specification and total training compute time per model; (5) preprocessing code for pose extraction and bounding-box tracking; and (6) evaluation scripts implementing the metrics defined in Section 5.3, to allow independent verification of any reported numbers against the raw model outputs.

This checklist is included specifically so that, once Sections 6 and 7 are populated with real measurements, readers and reviewers can verify that every reported number traces back to a logged, re-runnable experiment rather than an unverified estimate.